

THE DISCOVERY PILOT

Amiga Coverdisk Emulation & Videogaming the Incidental

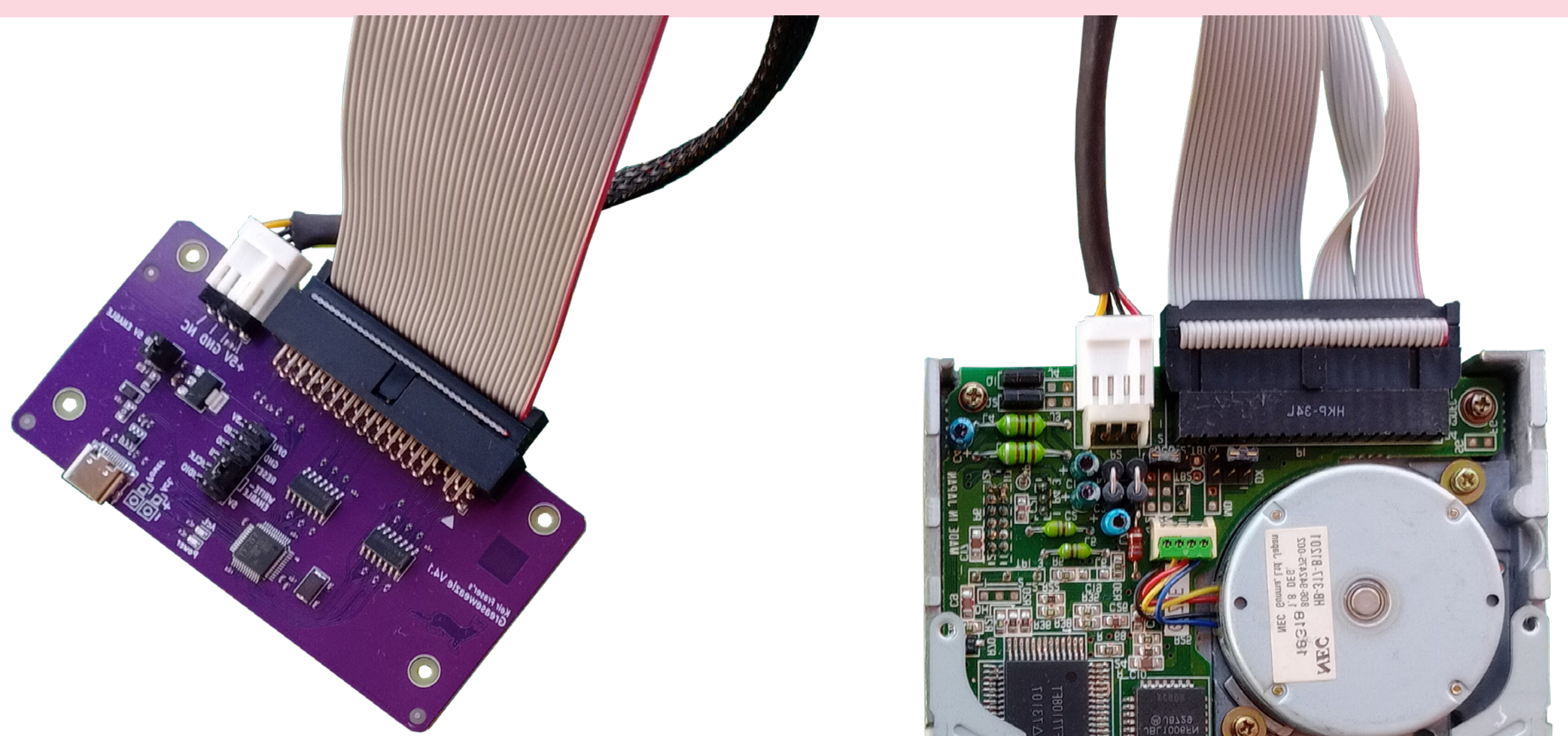
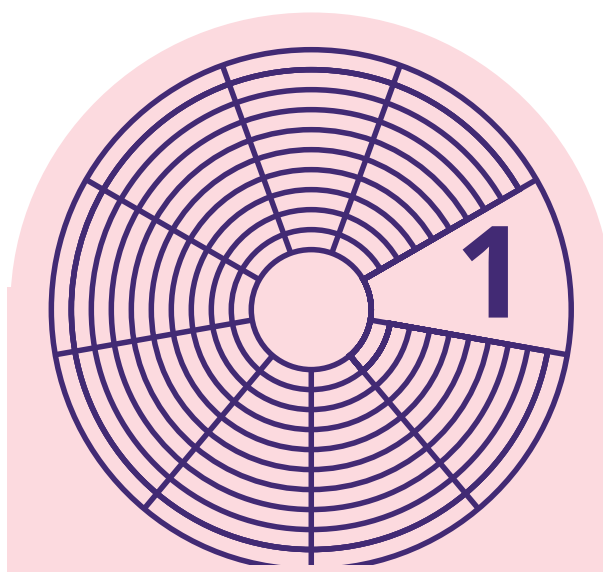


Fig 1. Disk capture equipment, featuring a Greaseweazle floppy disk controller connected via 34-pin ribbon cable to a Shugart-interface FDD, shown with two Amiga coverdisks [1], [2].



Coverdisk Migration

Migration hardware and software were selected for open source portability, ensuring workflows can be deployed seamlessly across library sites in Edinburgh and Glasgow. This prioritises accessibility for staff without specialist equipment.

Disk capture requires a standard Shugart-interface HD 3.5-inch floppy disk drive connected via 34-pin ribbon cable to a Greaseweazle v4.1 controller. The migration process generates four core assets:

- **Digital Preservation Master:** Flux-level stream files (.scp / SuperCard Pro format)
- **Access Asset:** Emulation derivatives (.rp9) for playback in Amiga Forever
- **Visual Record:** High-resolution floppy disk scans, edited in GIMP
- **Discovery Metadata:** Disk-level MARC records (itemising serials is out-of-scope)

The primary capture interface is FluxMyFloppy—a lightweight open-source GUI to generate Greaseweazle command-line sequences. For digital forensics and diagnostic analysis, HxC Floppy Emulator provides visualisations of magnetic-substrate damage and degradation at both the sector and track levels.

AMIGA COVERDISK PRESERVATION-TO-ACCESS WORKFLOW

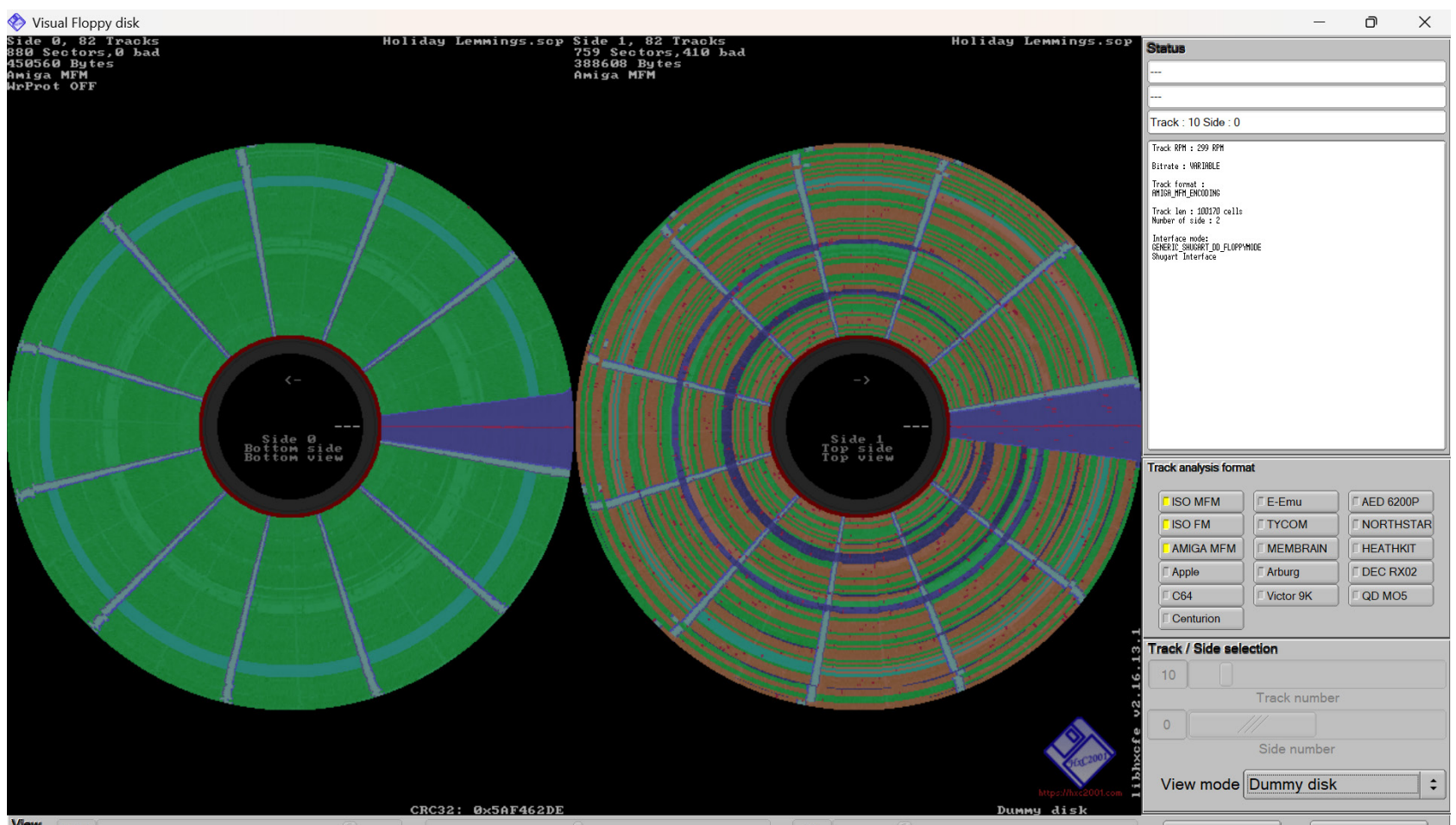
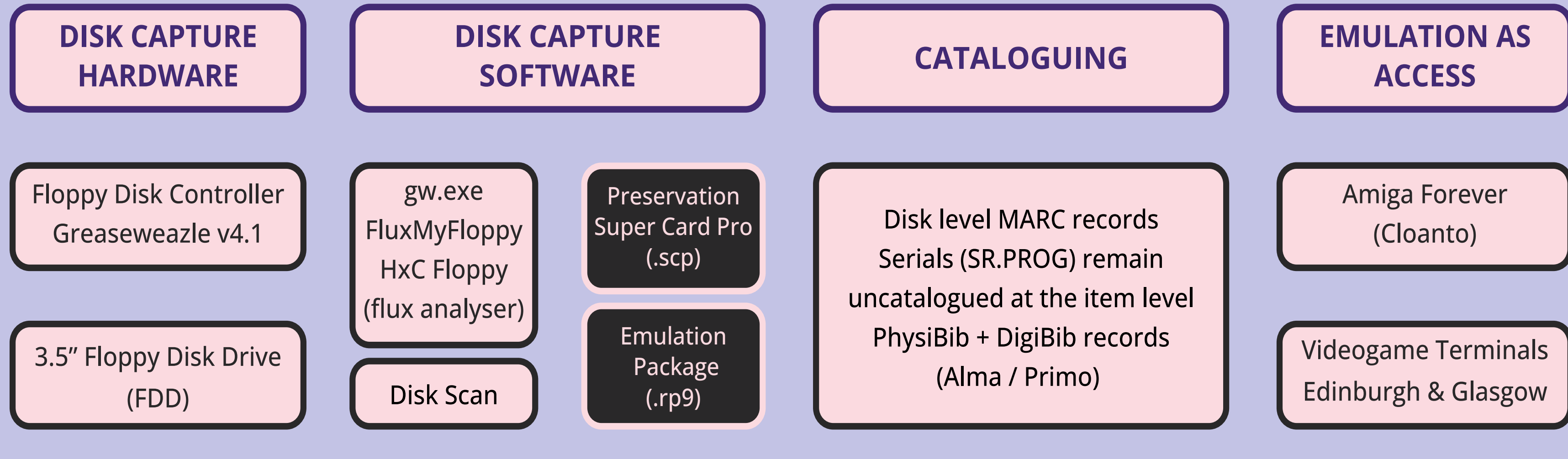


Fig 2. HxC Floppy Emulator flux-stream visual analyser revealing 410/759 bad sectors; the Side 1 concentric circles, depicted in red, likely confirm debris contamination, fungal growth, and/or binder hydrolysis (i.e., sticky-shed syndrome) [3].

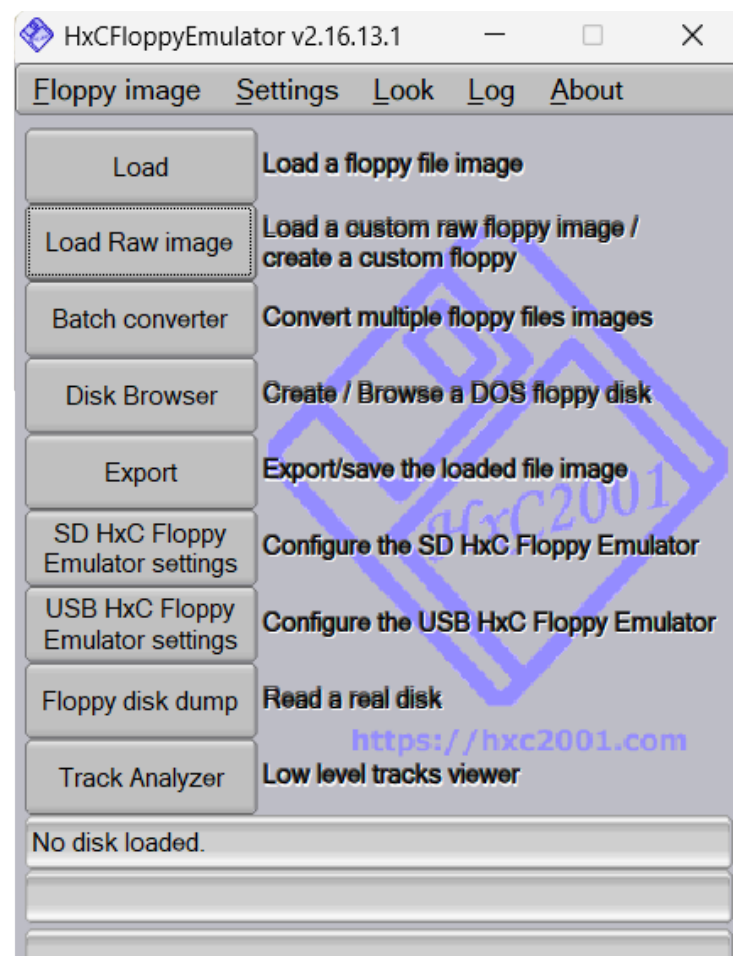


Fig 3. Discovery contributed to the Future Nostalgia Projects "Copy That Floppy!" guide [4].

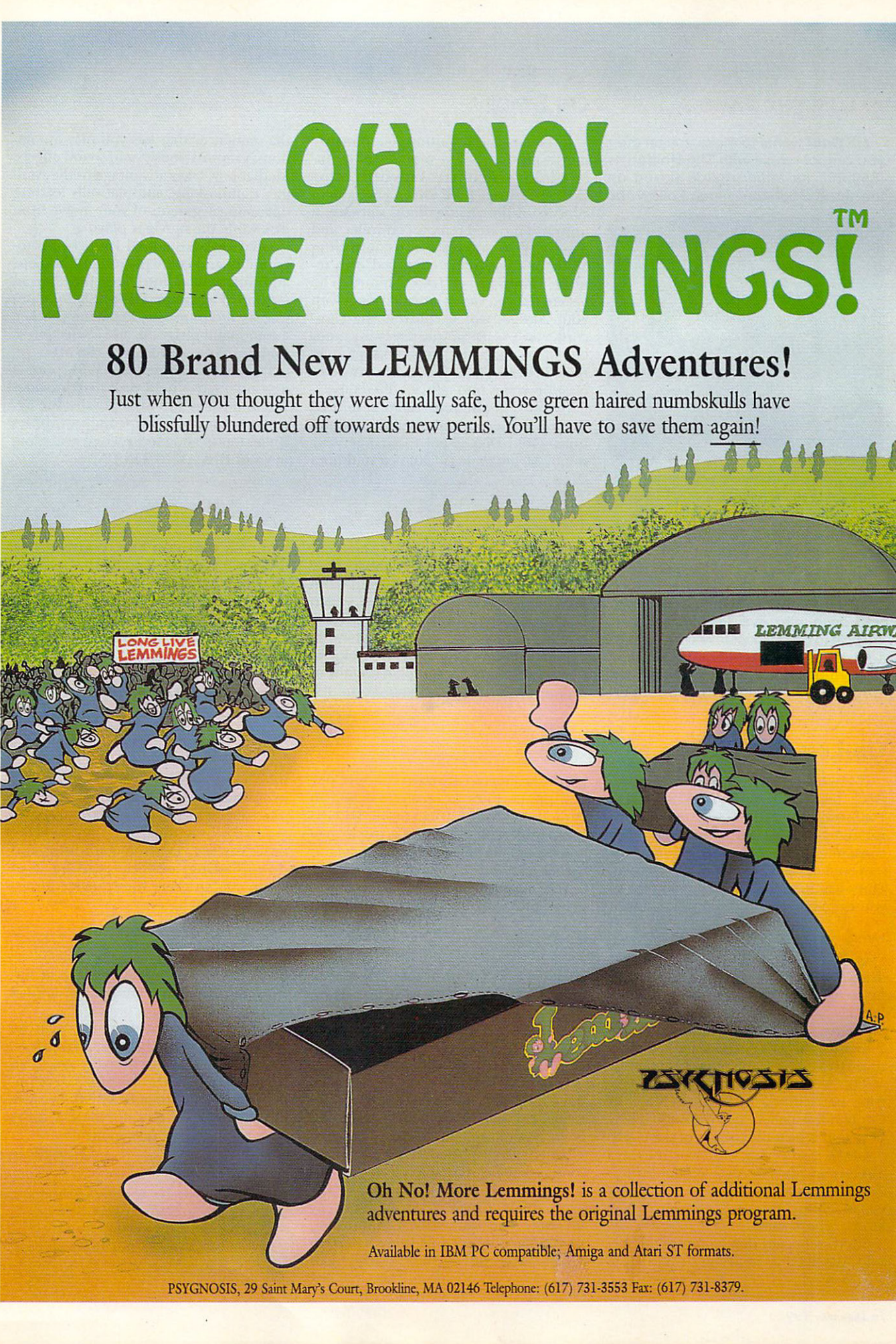


Fig 4. (L) Oh No! More Lemmings! expansion pack advertisement by DMA Design/Pygnosis [4]; (R) Front cover of Amiga Format, Issue 14 [5].



Fig 5. NLS Preservation Services 3.5-inch floppy disk box, piloted by Discovery.

Conor Q. Walker
National Library of Scotland
Discovery Project Officer

Digital Preservation Team
lee.hibberd@nls.uk
graham.purnell@nls.uk
conor.walker@nls.uk



Abstract

The National Library of Scotland's Discovery Pilot (2025–2027) is a two-year project addressing a pressing preservation horizon: the obsolescence of disk-based collections across CD Audio, DVD Video, and 3.5-inch floppy disk. Framed by the emergent digital—the transitional period from magnetic to optical media (1970s–1990s)—this poster highlights Amiga 3.5-inch coverdisks, which constitute the Library's largest computer game collection, acquired incidentally as supplemental material to legally deposited programming serials, from titles such as *Amiga Format* and *ST Amiga*.

Preservation-to-Access Cross-Sections:

- 1. Coverdisk Migration:** Approachable open source workflows utilising the Greaseweazle floppy disk controller to capture flux-level magnetic streams for playback within licensed Amiga Forever emulation environments.
- 2. Playing the Archive:** Deploying public emulation terminals in reading rooms in Edinburgh (George IV) and Glasgow (Kelvinhall), realising preservation brought to access.
- 3. Access Challenges & Cultural Porosity:** Addressing UK Legal Deposit gaps where videogames arrive incidentally, while co-leading a national network with National Museums Scotland (NMS) and the National Videogame Museum (NVM) to establish a shared framework for Scotland's videogaming legacy.

Conference Topic: Community

Keywords: Audiovisual Content | Emulation | Migration | Videogames | Floppy Coverdisks

Funding: National Library of Scotland Foundation (2025–2027)



Emulated Videogames & Software in Amiga Forever



Fig 6. Selected Amiga 3.5-inch coverdisk environments preserved during the Discovery Pilot, illustrating the range of videogames, time-based audio-visual demos, utility software, and Workbench OS interfaces captured for public emulation [7]–[15].

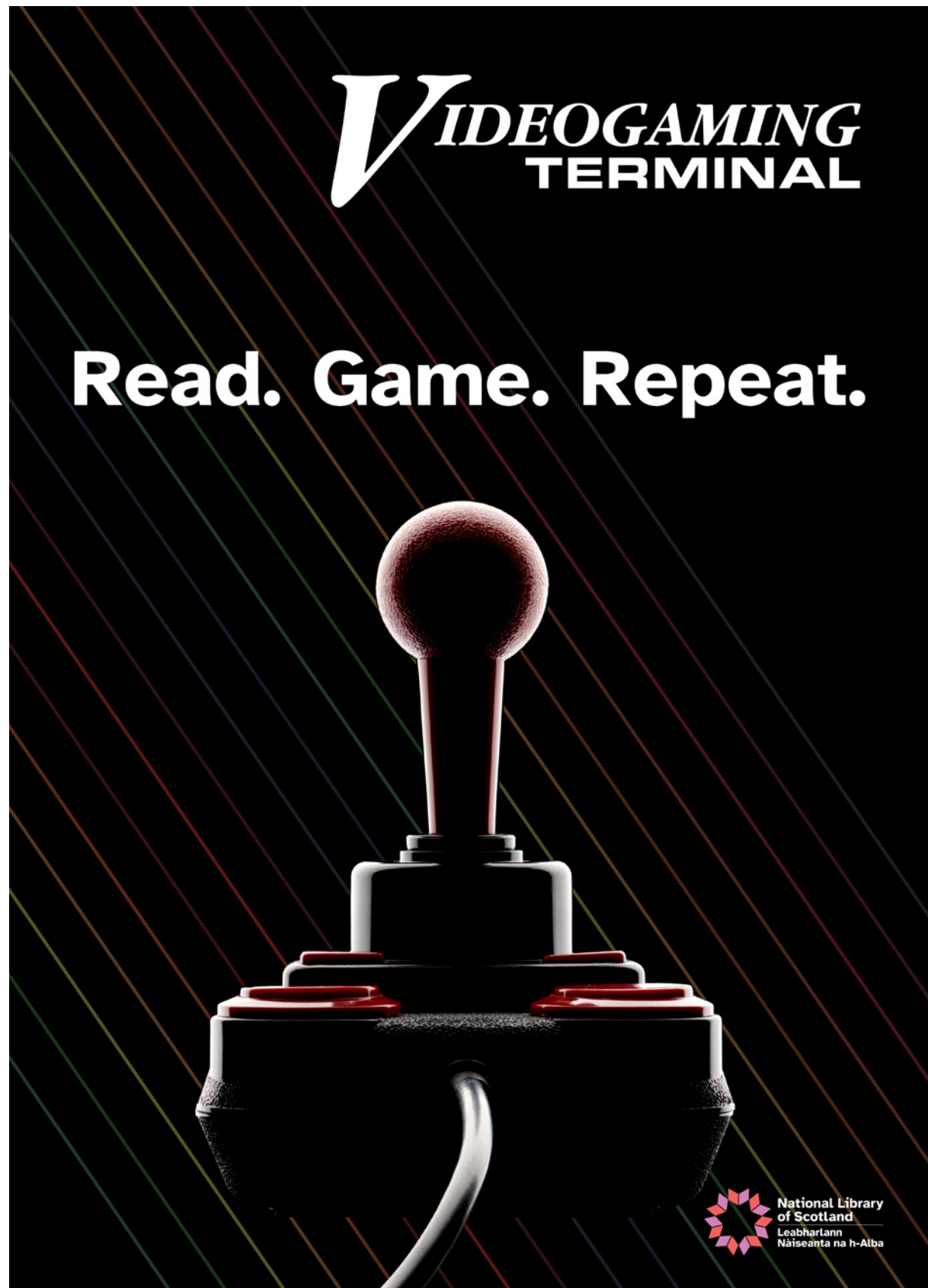


Fig 7. Videogaming Terminal poster by Callum Olsson [16].

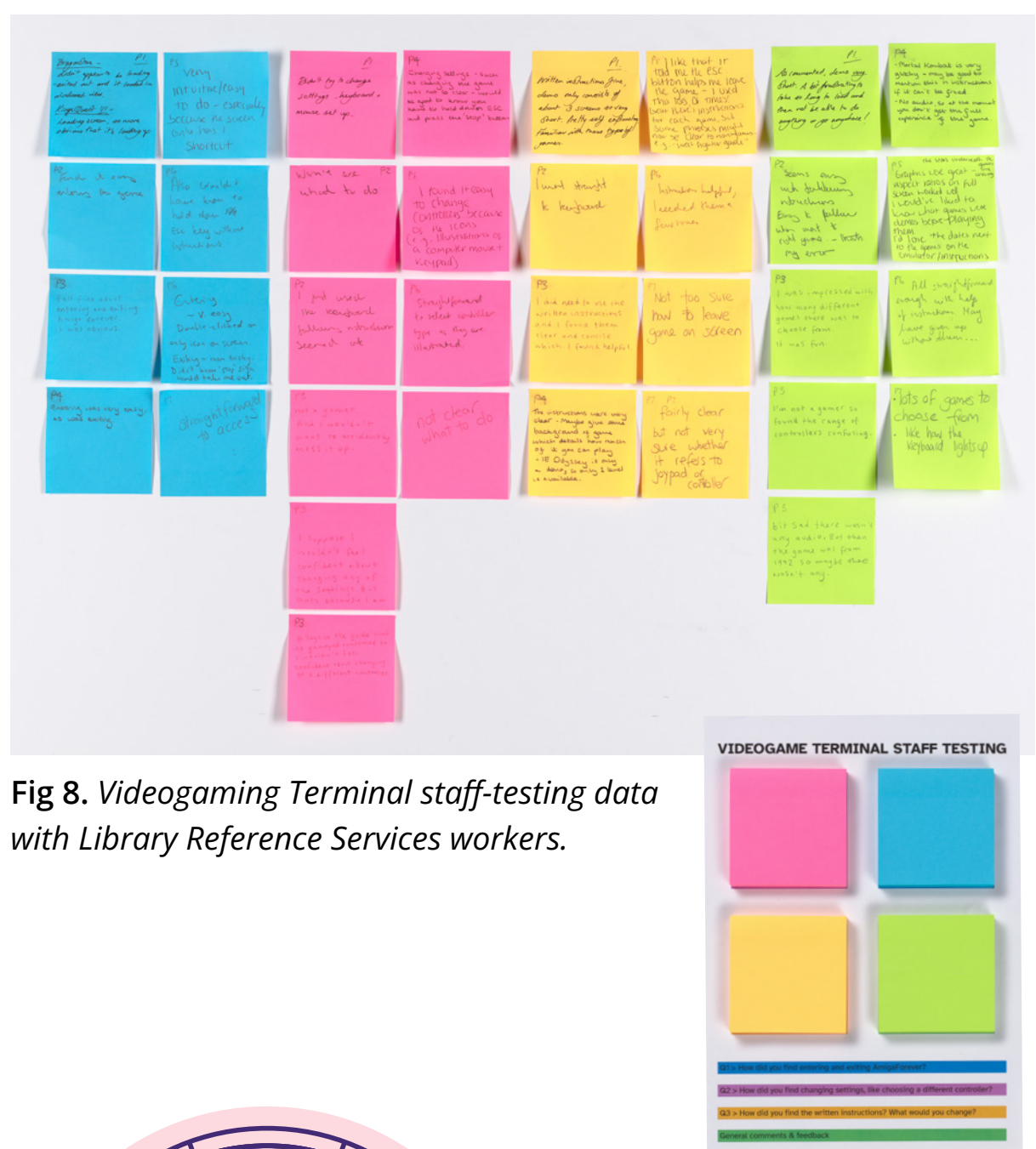


Fig 8. Videogaming Terminal staff-testing data with Library Reference Services workers.

Tomb Raiding Cinema

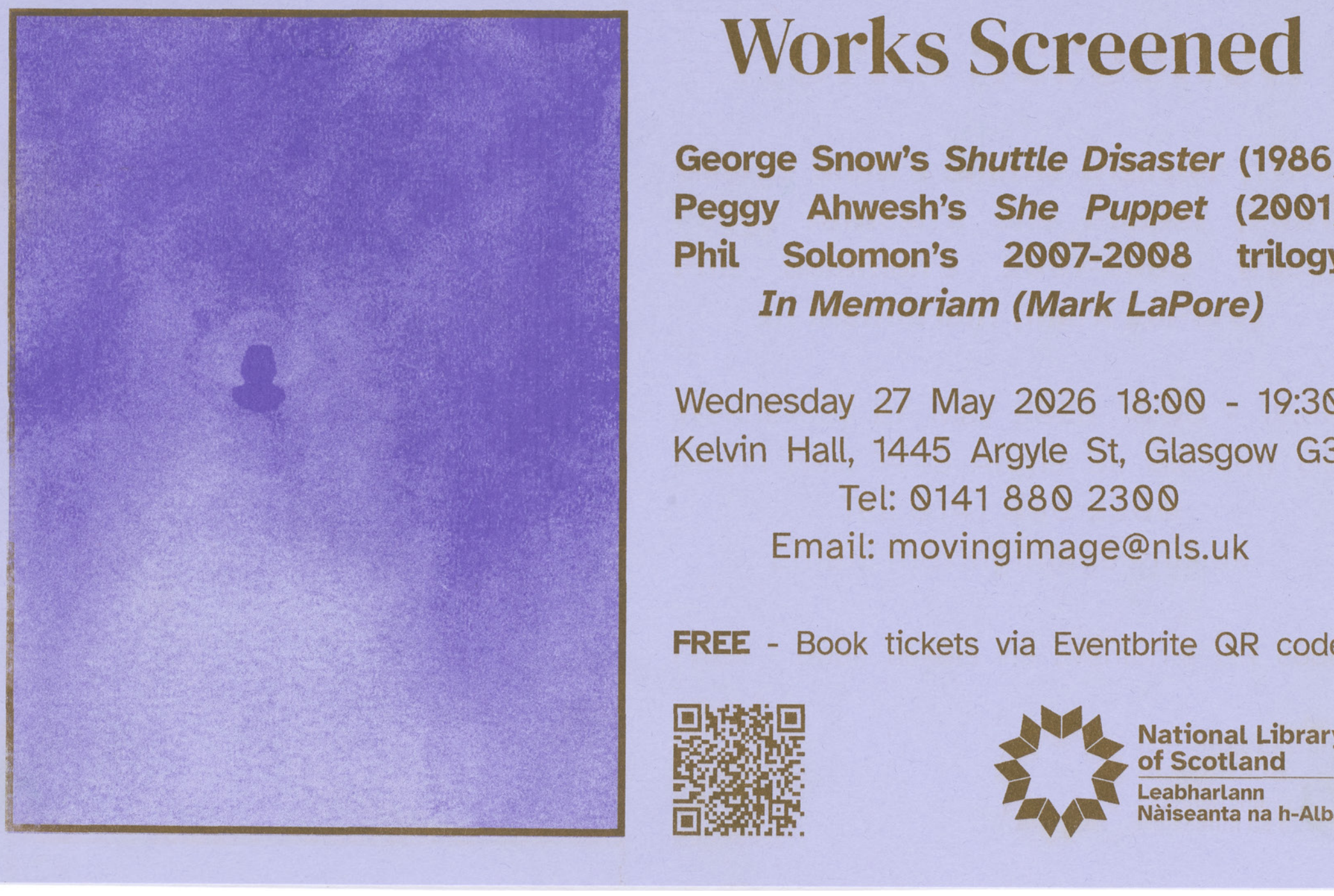
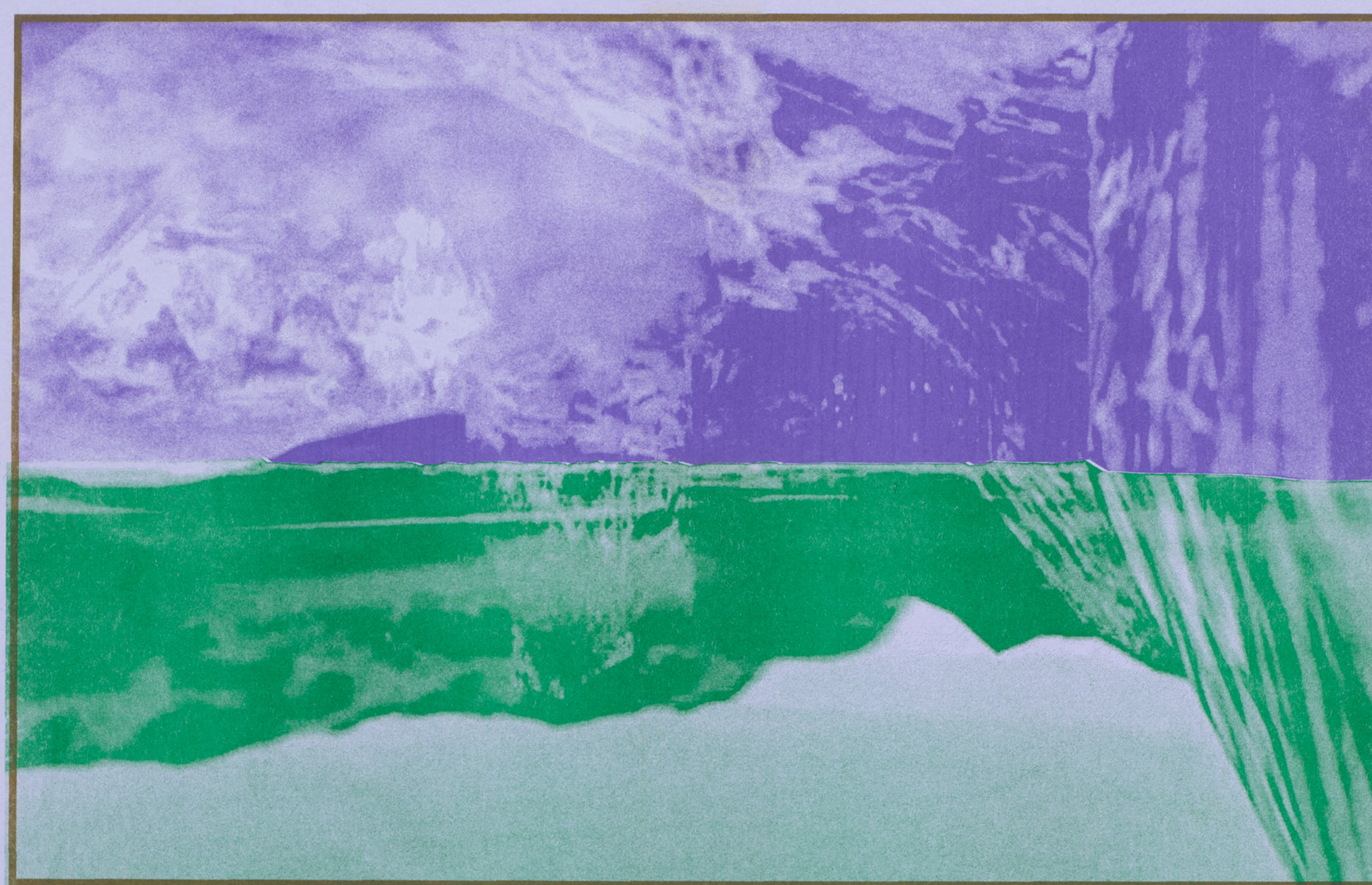
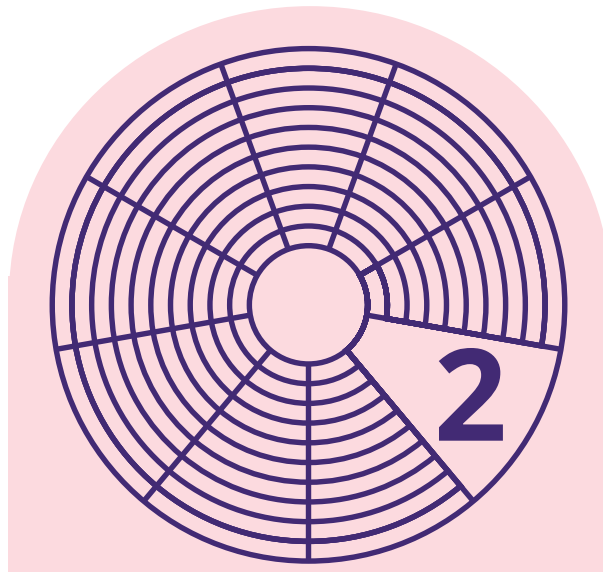


Fig 9. Risograph poster for the Discovery film programme "Tomb Raiding Cinema: Scratch video to experimental machinima" [17].



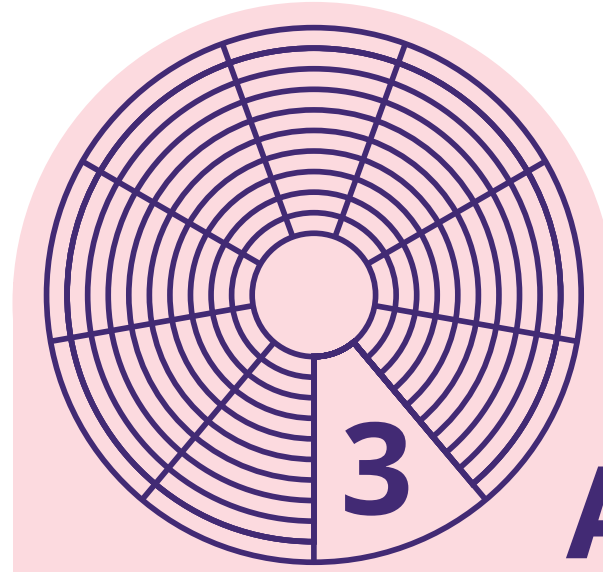
Playing the Archive

Installing dedicated videogaming stations in a library setting traditionally reserved for research re-orientes the public access offer toward interactive, time-based digital culture. Marking the first time videogames have been preserved by the Library, *Discovery* bridges long-term digital preservation of obsolescent magnetic carriers with immediate public access. Known as the **Videogame Terminal**, dedicated emulation stations are being deployed across two sites:

- **Edinburgh:** George IV Reading Room
- **Glasgow:** Moving Image, Music & Sound Collections Access Centre (Kelvinhall)

A primary interest of this setup lies in unlocking collection material that was not only previously inaccessible, but collected incidentally. Placing these terminals in research reading rooms invites new audiences into spaces traditionally reserved for text-based study, allowing visitors to explore software that would otherwise be lost to a magazine freebie. These coverdisks preserve a rich ecosystem of early digital culture: game demos, software in development, and work from emerging designers trying to break through. Beyond gaming, the Amiga was renowned as a system for digital art and audiovisual multimedia; the preserved disks capture early digital audio workstations (DAWs), sequencers, paint programmes, and demoscene culture.

On-site public access to this collection material is delivered through Section 40B of the Copyright, Designs and Patents Act 1988 (the dedicated terminal exception). The hardware architecture is intentionally minimalist: a stripped-down, off-network Windows PC hosting only a desktop launcher for licensed Amiga Forever emulation.



Access Challenges & Cultural Porosity

Videogames remain outside of the UK Non-Print Legal Deposit (NPLD) legislation, yet the Library has accumulated its largest videogame collection incidentally through coverdisks attached to legally deposited magazines. This inversion—preserving the incidental before the text—highlights the urgent need for flexible, sector-wide collecting mandates. No single institution can address these challenges alone; the Library is co-leading a national network with National Museums Scotland (NMS) and the National Videogame Museum (NVM) to establish a shared framework for collecting, preserving, and presenting Scotland's videogaming legacy. This collaboration culminated in *Playable Heritage: Creating a Collaborative Framework for Collecting Video Game Culture*, a two-day conference at NMS and the Library in Edinburgh combining expertise across the cultural memory sector, educators, and the videogame industry.

While the primary focus of Discovery remains the physical preservation and emulation of disk-based media, the project revealed how videogame culture inevitably seeps beyond playing and emulating games. Videogames bleed into wider artistic expression—including art, music, and experimental cinema such as machinima, where game engines supplant the traditional camera. Recognising these expressive afterlives is a key supplemental realisation of the pilot, pointing to rich avenues of cultural porosity worth deeper investigation in future projects.

References

- [1] "The One Amiga Coverdisk 49b," *The One Amiga*, no. 49, Disk B (K.G.B. game demo), Oct. 1992. London, U.K.: EMAP Images.
- [2] "CU Amiga Coverdisk 167," *CU Amiga*, Disk 167 (TB 303 Emulator), Oct. 1997. London, U.K.: EMAP Business Publishing.
- [3] "The One Amiga Coverdisk 76b," *The One Amiga*, no. 76, Disk B (*Holiday Lemmings* game demo), Jan. 1995. London, U.K.: EMAP Images.
- [4] L. Talboom et al., "Copy That Floppy: Imaging floppy disks for long-term preservation," University of Cambridge Library, 2025. [Online]. Available: [https://www.digipres.org/the-floppy-guide/].

- [5] "Oh No! More Lemmings!" advertisement, *Amiga World*, vol. 7, no. 12, p. 17, Dec. 1991. Peterborough, NH, USA: IDG Communications. [Periodical page, National Library of Scotland].
- [6] *Amiga Format*, no. 14, front cover, Sep. 1990. Bath, U.K.: Future Publishing. [Periodical cover, National Library of Scotland].
- [7] "Amiga Format Coverdisk 56b," *Amiga Format*, no. 56, Disk B (*Tornado* game demo), Feb. 1994. Bath, U.K.: Future Publishing. [3.5-inch floppy disk, National Library of Scotland].
- [8] *Nuclear War*, New World Computing, 1989. Published by U.S. Gold. [Software].
- [9] "Amiga Format Coverdisk 75b," *Amiga Format*, no. 75, Disk B (*Odyssey* game demo), Sep. 1995. Bath, U.K.: Future Publishing. [3.5-inch floppy disk, National Library of Scotland].
- [10] "ST Amiga Format Coverdisk 11," *ST Amiga Format*, no. 11, Disk 11 (*Blood Money* game demo), May 1989. Bath, U.K.: Future Publishing. [3.5-inch floppy disk, National Library of Scotland].
- [11] "Amiga Format Coverdisk 96a," *Amiga Format*, no. 96, Disk A (*Personal Paint* v6.4), Apr. 1997. Bath, U.K.: Future Publishing.

- [12] *Amiga Format*, no. 50, Disk C (*Dozer* demo / Workbench OS window), Apr. 1994. Bath, U.K.: Future Publishing. [3.5-inch floppy disk, National Library of Scotland].
- [13] "CU Amiga Coverdisk 41," *CU Amiga*, Disk 41 (*Apod*, *Catch Em*, and *QBIC* game demos / Workbench OS window), Sep. 1992. London, U.K.: EMAP Business Publishing. [3.5-inch floppy disk, National Library of Scotland].
- [14] "Amiga Format Coverdisk 30b," *Amiga Format*, no. 30, Disk B (*Sequencer One* full software), Jan. 1992. Bath, U.K.: Future Publishing. [3.5-inch floppy disk, National Library of Scotland].
- [15] *Amiga Format*, no. 96, Disk A (*Personal Paint* v6.4), Apr. 1997. Bath, U.K.: Future Publishing.
- [16] C. Olsson, *Videogaming Terminal*, 2026. [Poster]. National Library of Scotland Imaging Services, Edinburgh, U.K.
- [17] C. Q. Walker, *Tomb Raiding Cinema: Scratch video to experimental machinima*, May 27, 2026. [Risograph poster]. National Library of Scotland, Moving Image, Music & Sound Collections at Kelvin Hall, Glasgow, U.K.